

GRADIFY: AN AI-DRIVEN CAREER COUNSELING AND CAREER COACHING PLATFORM

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ABSTRACT

Choosing the right technical domain is a critical decision for Computer Engineering students, as it impacts their academic focus, projects, and career opportunities. Traditional career guidance systems are often static and non-adaptive, leading to misaligned skill development and reduced employability. This paper proposes Gradify, an AI-driven platform that leverages the Simplified Adaptive Career Recommendation Algorithm (SACRA) to provide personalized, explainable, and adaptive domain recommendations. Gradify continuously updates guidance based on academic performance, skill assessments, and learning progress, while offering domain-specific roadmaps, project suggestions, and career preparation tools. The system aims to improve decision-making, skill alignment, and employability outcomes for students in computer engineering programs.

INTRODUCTION

Computer Engineering students often struggle to select a suitable technical domain, such as Artificial Intelligence, Web Development, Data Science, or Cybersecurity. Poor domain choices—often influenced by peer pressure or incomplete self-assessment—can lead to misaligned skills and reduced career opportunities.

Traditional guidance systems are generic, static, and do not account for continuous learning progress. To address this gap, Gradify provides adaptive, explainable, and data-driven recommendations using the SACRA algorithm. The system continuously updates suggestions based on student performance, interests, and career goals, helping students make informed specialization decisions.

RELATED WORK

Several AI-driven career counseling systems have been proposed in recent years. Agrawal *et al.* presented an AI-based career counseling platform that analyzes student interests and academic performance to recommend suitable career paths. However, the system

lacks continuous adaptability and domain-specific specialization.

Birajdar *et al.* emphasized the use of psychometric tests and predictive analytics for career guidance, but their approach does not consider real-time performance tracking or technical skill evaluation. COUNSEL-BOT introduced chatbot-based career mapping but suffers from limited scalability and lack of adaptive intelligence.

Other studies have explored AI-based resume analysis and course recommendation systems; however, these tools focus on isolated components rather than offering an integrated, adaptive platform. Most existing systems act as static recommenders, failing to evolve as students acquire new skills.

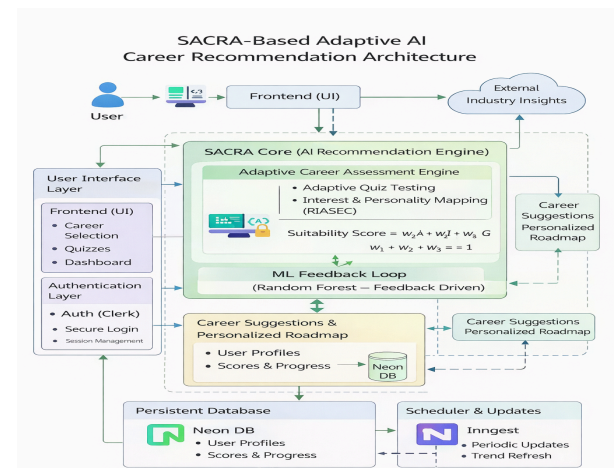
Research Gap Identified:

There is a lack of adaptive, explainable, and domain-specific AI systems that continuously guide Computer Engineering students in selecting and refining their technical specialization.

PROPOSED METHODOLOGY

This section describes the architecture, data processing, algorithmic framework, and adaptive feedback mechanism used in Gradify.

A. System Architecture



The proposed system architecture consists of five major layers:

- i) User Interface Layer – Web-based interface for quizzes, dashboards, roadmaps, and progress tracking.
- ii) Authentication Layer – Secure login and profile management.
- iii) Data Processing Layer – Preprocessing of academic, skill, and interest data.
- iv) AI Recommendation Engine –Core SACRA algorithm.
- v) Resource & Feedback Layer – Roadmaps, assessments, and adaptive updates.

This modular architecture ensures scalability, maintainability, and real-time adaptability.

B. Data Acquisition and Preprocessing

Gradify collects structured and unstructured data from students and represents it as a feature vector:

$$U = \{A, I, S, G\}$$

Where:

A : Academic performance in core CS subjects (DSA, OOP, DBMS, OS).

I : Interest scores from ML-powered quizzes.

S : Practical skill proficiency (coding tests, projects).

G : Career goals (AI Engineer, Full Stack Developer, etc.)

All values are normalized using Min-Max normalization:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

C. The SACRA Algorithm

SACRA (Simplified Adaptive Career Recommendation Algorithm) is designed to be lightweight, explainable, and adaptive.

Step 1: Domain Weight Calculation

For each domain (d):

$$Score_d = w_1A + w_2I + w_3S + w_4G$$

Where, $w_1 + w_2 + w_3 + w_4 = 1$

Step 2: Domain Ranking

Domains are ranked in descending order based on their $Score_d$.

Step 3: Recommendation Output

Top-ranked domain(s) are recommended along with:

- i) Skill roadmap
- ii) Project suggestions
- iii) Learning timeline

D. Adaptive Feedback Mechanism

SACRA dynamically updates recommendations using performance feedback:

$$Score_d^{t+1} = Score_d^t + \alpha(P_t - P_{t-1})$$

Where:

P_t : Current performance score at time .

P_{t-1} : Previous performance score.

α (Alpha) : Learning rate (determines how quickly the system reacts to new data).

This ensures recommendations evolve with student growth.

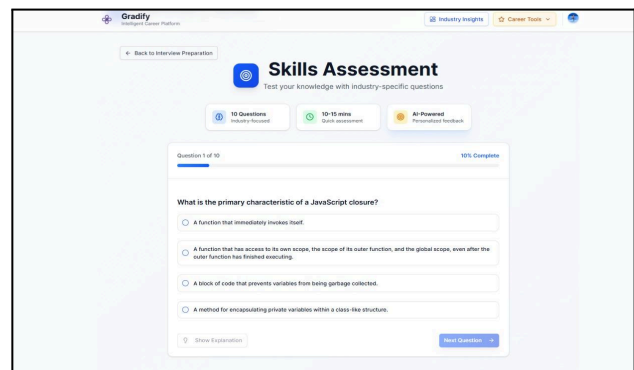
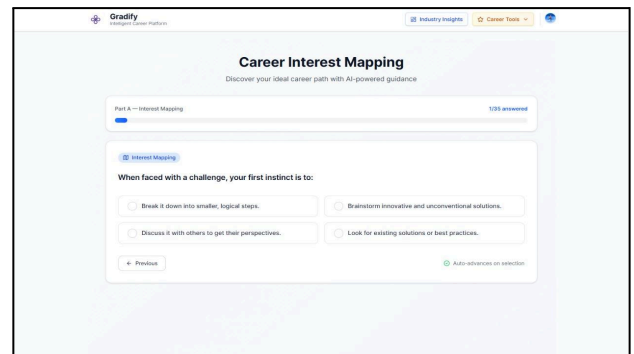
SYSTEM IMPLEMENTATION AND WORKFLOW

A. Functional Modules

The Gradify platform consists of the following key functional modules, each designed to provide a seamless and adaptive learning experience for Computer Engineering students:

i) Domain Identification Quiz

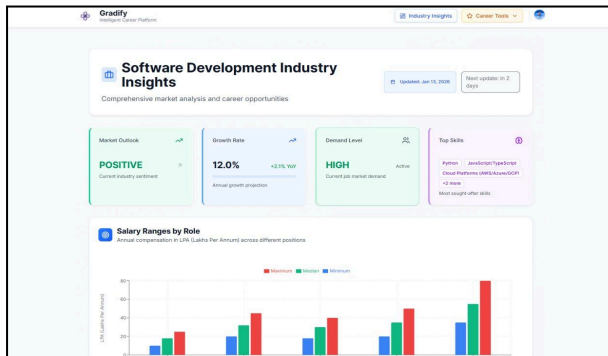
This module evaluates a student's interests, aptitude, and knowledge in various computer engineering domains through adaptive quizzes. The results act as initial input for the SACRA algorithm to generate personalized recommendations.



ii) Industry Insight and Weekly Trend Analysis

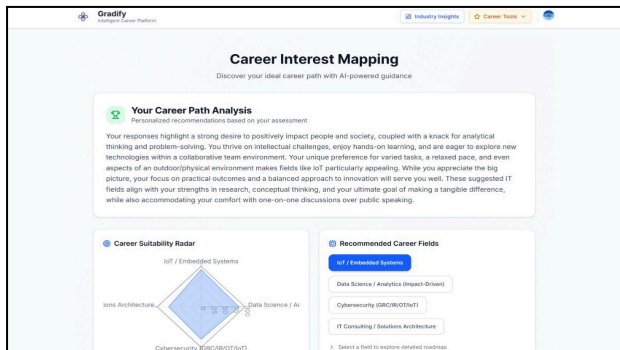
After identifying the preferred domain, this module

continuously analyzes current industry trends, emerging technologies, and in-demand skills relevant to the selected domain. The system fetches and updates industry insights on a weekly basis, ensuring that students remain aligned with real-world market requirements. Based on these insights, learners are informed about evolving skill demands, recommended technologies to focus on, and shifting industry expectations, thereby enabling informed and future-ready career decisions.



iii) Adaptive Learning Roadmap Generator

Based on the quiz results and skill gap analysis, this module creates a personalized, step-by-step learning roadmap. It specifies which skills to acquire first, recommended resources, and milestones for progress tracking.



iv) Resume and Email Generator

This module helps students prepare professional resumes and cover letters tailored to their chosen domain. It also allows bulk email generation for internship and job applications.

Resume Builder
Create a professional resume that stands out

Resume Progress: 1 of 4 steps

Build Your Resume
Fill in your details to create a professional resume

Personal Information
Basic contact details and professional links

Full Name: John Doe
Email: john@example.com
Phone: +1 (555) 123-4567
LinkedIn: https://linkedin.com/in/username
GitHub: https://github.com/username

Cover Letters
Manage and track your professional cover letters

Senior Software Engineer
50%
Created: 12/12/2023

Bulk Email Campaign
Reach multiple recipients efficiently with our bulk email tool

Send your cover letters to multiple HR professionals and recruiters at once. Perfect for mass applications or networking campaigns.

- Send to multiple recipients
- Track email delivery status
- Personalized email templates

[Open Bulk Email Sender](#)

Create Cover Letter
Let AI craft a personalized cover letter for your dream job

Job Details
Provide the job information and we'll generate a professional cover letter

Job Title
e.g. Senior Software Engineer, Product Manager, Marketing Specialist

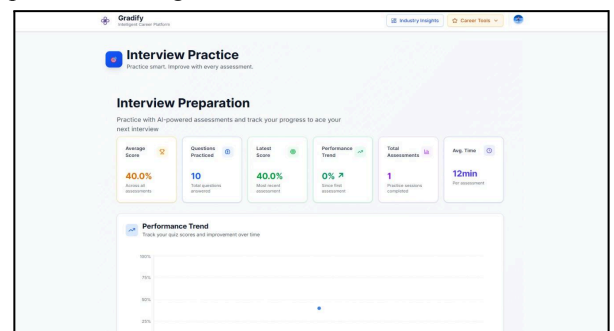
Company Name
e.g. Google, Microsoft, Amazon

Job Description
Paste the complete job description here. Include key responsibilities, requirements, and any specific qualifications mentioned.

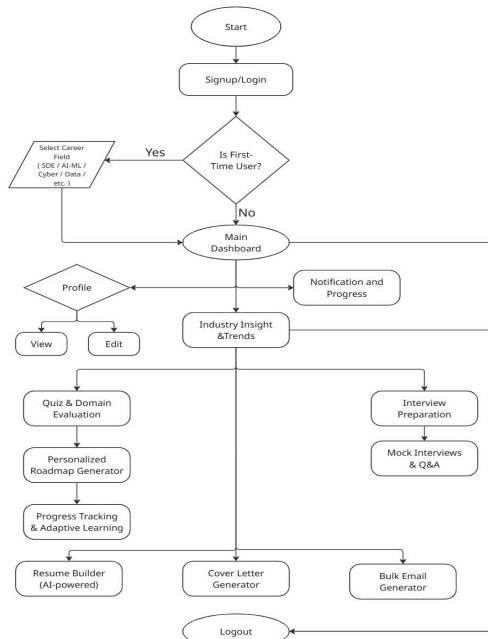
[Generate Cover Letter](#)

v) Interview Preparation Module

Focused on domain-specific interview readiness, this module provides practice questions, mock interviews, and scenario-based exercises to improve technical and problem-solving skills.



B. Operational Workflow (According to Proposed System Workflow Diagram)



The operational workflow of the Gradify platform follows a structured and user-centric sequence, as illustrated in the workflow diagram. The system is designed to guide Computer Engineering students step-by-step from onboarding to domain selection, skill development, and career preparation.

Step 1: System Initialization

The workflow begins when the user launches the Gradify platform. The system initializes the user interface and prepares backend services required for authentication, assessment, and recommendation processing.

Step 2: User Signup / Login

The user is prompted to either sign up or log in to the system.

i) Signup:

New users register by providing basic details such as name, email ID, academic background, and preferred learning goals.

ii) Login:

Existing users authenticate using secure credentials. Authentication ensures personalized access to stored profiles, progress history, and recommendations.

Step 3: First-Time User Identification

After authentication, the system checks whether the user is accessing the platform for the first time.

i) If Yes (First-Time User):

The user is redirected to the Career Field Selection module.

ii) If No (Returning User):

The user is directly redirected to the Main Dashboard. This decision node allows the platform to customize the user journey based on experience level.

Step 4: Career Field Selection (First-Time Users)

First-time users are asked to select their preferred technical domain, such as:

i) Software Development (SD)

ii) Artificial Intelligence / Machine Learning (AI/ML)

iii) Data Science

iv) Cybersecurity

v) Other related computing domains

This selection acts as an initial preference input for the SACRA algorithm and helps in tailoring assessments and roadmaps.

Step 5: Main Dashboard Access

The Main Dashboard serves as the central navigation hub of the system. From this dashboard, users can access all core functionalities, including profile management, assessments, industry insights, progress tracking, and career preparation tools.

Step 6: Profile Management

Users can access their Profile section from the dashboard, which provides two options:

i) View Profile:

Displays personal information, selected domain, assessment history, and progress metrics.

ii) Edit Profile:

Allows users to update academic details, interests, skills, and career goals.

Profile updates dynamically influence future recommendations generated by SACRA.

Step 7: Industry Insights and Trends

The system provides an Industry Insight & Trends module that displays:

i) Emerging technologies

ii) Domain-specific skill demands

iii) Market trends relevant to the selected career field

This ensures students remain aligned with current industry expectations while planning their learning paths.

Step 8: Quiz and Domain Evaluation

Users proceed to the Quiz & Domain Evaluation module, where adaptive assessments are conducted to evaluate:

- i) Core subject understanding
- ii) Domain-specific aptitude
- iii) Interest consistency

The results of this evaluation serve as key inputs ((A, I, S)) to the SACRA algorithm.

Step 9: Personalized Roadmap Generation

Based on quiz results and user profile data, the system generates a Personalized Learning Roadmap, which includes:

- i) Skill-wise learning sequence
- ii) Recommended resources
- iii) Timeline-based milestones

This roadmap is uniquely generated for each user and adapts over time.

Step 10: Progress Tracking and Adaptive Learning

As users follow the roadmap:

- i) Their progress is continuously monitored
- ii) Performance data is recorded
- iii) Learning patterns are analyzed

The adaptive learning mechanism ensures that the roadmap and domain recommendations evolve as the user improves.

Step 11: Career Preparation Modules

The system supports career readiness through additional modules accessible from the dashboard:

- i) Resume Builder (AI-Powered):

Generates domain-specific resumes based on user skills and progress.

- ii) Cover Letter Generator:

Creates personalized cover letters aligned with selected domains.

- iii) Bulk Email Generator:

Assists users in applying for internships and opportunities efficiently.

Step 12: Interview Preparation

The Interview Preparation module helps users prepare for technical interviews through:

- i) Concept revision
- ii) Domain-specific questions
- iii) Practice scenarios

Step 13: Mock Interviews and Q&A

Users can participate in Mock Interviews & Q&A sessions, which simulate real interview environments and provide feedback for improvement.

Step 14: Notifications and Progress Updates

Throughout the workflow, the system sends notifications and progress updates to:

- i) Remind users of milestones
- ii) Highlight improvement areas
- iii) Encourage consistent learning

These notifications enhance engagement and motivation.

Step 15: Logout

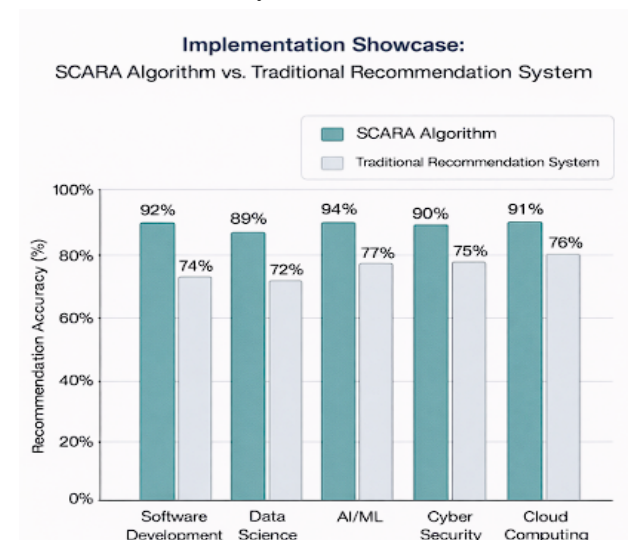
Once the session is complete, users can securely log out of the platform. All progress data is saved for future sessions, ensuring continuity.

V. RESULTS AND DISCUSSION

A. Experimental Setup

The system was tested using simulated student profiles representing different academic strengths and interests across CS domains.

B. Performance Analysis



Method	Adaptability	Explainability	Accuracy
Traditional Counseling	Low	High	Low
MLStatic Models	Medium	Low	Medium
SACRA (Proposed)	High	High	High

SACRA demonstrated improved recommendation consistency and alignment.

C. Discussion of Explainability

Unlike deep learning models, SACRA provides transparent scoring and clear reasoning behind recommendations, which is critical for student trust and academic decision-making.

VI. CONCLUSION AND FUTURE WORK

This paper presented Gradify, an AI-based career counseling platform powered by the SACRA algorithm. By combining simplicity, adaptability, and transparency, SACRA addresses critical gaps in existing career guidance systems. Future enhancements include integration of real-time job market APIs, emotional intelligence modeling, and large-scale real-world validation.

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